

Aurora SVMAT Metric Formula Reference

Main body: v2/v3 research proxies and v4 physical aperture metrics. Appendix: legacy engineering metrics.

This document records the explicit mathematical definitions currently implemented in Aurora SVMAT Lab. The notation is aligned to the control-point sequence view $c_i = \{\theta_i, w_i, M_i, z_i\}$.

RESEARCH USE ONLY. This document reflects the current code implementation and should be interpreted as a research reference rather than a clinical specification.

Notation

i indexes adjacent control-point intervals from control point i to $i+1$.

θ_i is the gantry angle in degrees at control point i .

z_i is the longitudinal or axial position in millimeters at control point i .

w_i is the cumulative meterset weight at control point i .

For V2/V3/legacy only, A_i is the scalar aperture width surrogate at control point i , defined as the summed positive opening across both MLC layers.

L_i is the total leaf-position vector at control point i across both layers.

$\Delta \theta_i = \text{wrapped}(\theta_{i+1} - \theta_i)$, constrained to $[-180, 180]$ degrees.

$\Delta z_i = z_{i+1} - z_i$.

$\Delta w_i = w_{i+1} - w_i$.

$\Delta A_i = |A_{i+1} - A_i|$.

ΔL_i = summed absolute leaf travel between adjacent control points across both layers.

$CV(x) = \text{std_pop}(x) / \text{mean}(x)$, evaluated only on valid non-empty interval sets.

Section 1. V2 Metrics

longitudinal_travel_mm

Total absolute longitudinal travel

$\text{longitudinal_travel_mm} = \sum_i |\Delta z_i|$

Meaning: Total absolute longitudinal travel reconstructed from adjacent control-point isocenter z positions.

total_rotation_deg

Total absolute gantry rotation

$\text{total_rotation_deg} = \sum_i |\Delta \theta_i|$

Meaning: Total absolute gantry rotation reconstructed from the observed angle sequence rather than declared direction tags.

rotations

Total gantry rotations

```
rotations = total_rotation_deg / 360
```

Meaning: Total gantry rotation expressed in full 360-degree turns.

travel_per_rotation_mm

Longitudinal travel per full gantry rotation

```
travel_per_rotation_mm = longitudinal_travel_mm / rotations
```

Meaning: Average longitudinal travel per gantry rotation, derived from total travel and total rotation.

projection_pitch_mean

Mean projection pitch

```
projection_pitch_i = |delta z_i| / |delta theta_i|  
projection_pitch_mean = mean_i(projection_pitch_i)
```

Meaning: Mean projection pitch, defined as the mean of $|\text{delta } z| / |\text{delta } \theta|$ over valid adjacent control-point intervals.

projection_pitch_cv

Projection pitch variability

```
projection_pitch_i = |delta z_i| / |delta theta_i|  
projection_pitch_cv = CV(projection_pitch_i)
```

Meaning: Coefficient of variation of projection pitch. Lower values mean steadier theta-z coupling.

projection_mu_density_mean_proxy

Mean projection meterset-weight density proxy

```
projection_mu_density_proxy_i = delta w_i / |delta z_i|  
projection_mu_density_mean_proxy = mean_i(projection_mu_density_proxy_i)
```

Meaning: Mean projection MU-density proxy, using cumulative meterset-weight change per millimeter of longitudinal travel.

projection_mu_density_cv_proxy

Variability of the projection meterset-weight density proxy

```
projection_mu_density_proxy_i = delta w_i / |delta z_i|  
projection_mu_density_cv_proxy = CV(projection_mu_density_proxy_i)
```

Meaning: Coefficient of variation of the projection MU-density proxy across adjacent control-point intervals.

projection_aperture_change_mean

Mean aperture-change rate along the longitudinal axis

```
delta A_i = |A_(i+1) - A_i|  
projection_aperture_change_i = delta A_i / |delta z_i|  
projection_aperture_change_mean = mean_i(projection_aperture_change_i)
```

Meaning: Mean projection aperture-change rate, defined as absolute aperture-width change per millimeter of longitudinal travel.

projection_aperture_change_cv

Variability of the aperture-change rate

```
delta A_i = |A_(i+1) - A_i|  
projection_aperture_change_i = delta A_i / |delta z_i|  
projection_aperture_change_cv = CV(projection_aperture_change_i)
```

Meaning: Coefficient of variation of the projection aperture-change rate.

projection_leaf_travel_mean

Mean leaf-travel rate across both MLC layers

```
delta L_i = sum_leaves |leaf_(i+1) - leaf_i| across MLCX1 and MLCX2
projection_leaf_travel_i = delta L_i / |delta z_i|
projection_leaf_travel_mean = mean_i(projection_leaf_travel_i)
```

Meaning: Mean projection leaf-travel rate, defined as summed leaf travel per millimeter of longitudinal travel.

projection_leaf_travel_cv

Variability of the total leaf-travel rate

```
delta L_i = sum_leaves |leaf_(i+1) - leaf_i| across MLCX1 and MLCX2
projection_leaf_travel_i = delta L_i / |delta z_i|
projection_leaf_travel_cv = CV(projection_leaf_travel_i)
```

Meaning: Coefficient of variation of the projection leaf-travel rate.

projection_leaf_travel_mean_mlcx1

Mean leaf-travel rate for MLCX1

```
delta L_i^(MLCX1) = sum_x1 |x1_(i+1) - x1_i|
projection_leaf_travel_mlcx1_i = delta L_i^(MLCX1) / |delta z_i|
projection_leaf_travel_mean_mlcx1 = mean_i(projection_leaf_travel_mlcx1_i)
```

Meaning: Mean projection leaf-travel rate for the MLCX1 layer only, normalized by longitudinal travel.

projection_leaf_travel_cv_mlcx1

Variability of the MLCX1 leaf-travel rate

```
delta L_i^(MLCX1) = sum_x1 |x1_(i+1) - x1_i|
projection_leaf_travel_mlcx1_i = delta L_i^(MLCX1) / |delta z_i|
projection_leaf_travel_cv_mlcx1 = CV(projection_leaf_travel_mlcx1_i)
```

Meaning: Coefficient of variation of the MLCX1 projection leaf-travel rate.

projection_leaf_travel_mean_mlcx2

Mean leaf-travel rate for MLCX2

```
delta L_i^(MLCX2) = sum_x2 |x2_(i+1) - x2_i|
projection_leaf_travel_mlcx2_i = delta L_i^(MLCX2) / |delta z_i|
projection_leaf_travel_mean_mlcx2 = mean_i(projection_leaf_travel_mlcx2_i)
```

Meaning: Mean projection leaf-travel rate for the MLCX2 layer only, normalized by longitudinal travel.

projection_leaf_travel_cv_mlcx2

Variability of the MLCX2 leaf-travel rate

```
delta L_i^(MLCX2) = sum_x2 |x2_(i+1) - x2_i|
projection_leaf_travel_mlcx2_i = delta L_i^(MLCX2) / |delta z_i|
projection_leaf_travel_cv_mlcx2 = CV(projection_leaf_travel_mlcx2_i)
```

Meaning: Coefficient of variation of the MLCX2 projection leaf-travel rate.

theta_z_coupling_cv

Theta-z coupling variability

```
theta_z_coupling_cv = projection_pitch_cv
```

Meaning: Theta-z coupling variability, reported here as the same projection-pitch coefficient of variation.

mu_z_coupling_cv_proxy

Meterset-weight-density to z coupling variability proxy

`mu_z_coupling_cv_proxy = projection_mu_density_cv_proxy`

Meaning: MU-z coupling variability proxy, reported as the coefficient of variation of meterset-weight density over longitudinal travel.

mlc_z_coupling_cv

MLC-z coupling variability across both layers

`mlc_z_coupling_cv = projection_leaf_travel_cv`

Meaning: MLC-z coupling variability, reported here as the coefficient of variation of the projection leaf-travel rate.

mlcx1_z_coupling_cv

MLCX1-z coupling variability

`mlcx1_z_coupling_cv = projection_leaf_travel_cv_mlcx1`

Meaning: MLCX1-z coupling variability, reported as the coefficient of variation of the MLCX1 projection leaf-travel rate.

mlcx2_z_coupling_cv

MLCX2-z coupling variability

`mlcx2_z_coupling_cv = projection_leaf_travel_cv_mlcx2`

Meaning: MLCX2-z coupling variability, reported as the coefficient of variation of the MLCX2 projection leaf-travel rate.

Section 2. V3 Research Metrics

These research-facing v3 metrics are the immediately implementable Aurora extensions derived directly from currently parsed RTPLAN control-point data.

reversal_symmetry_index

Bidirectional modulation symmetry

```
mean over selected burden families of [1 - abs(forward_mean - backward_mean) /  
(abs(forward_mean) + abs(backward_mean))]
```

Meaning: Symmetry score comparing forward and backward beam groups across matched modulation summaries. Higher values mean more balanced bidirectional modulation.

forward_backward_metric_difference

Forward-versus-backward modulation difference

```
abs(mean_forward(coupled_modulation_index) - mean_backward(coupled_modulation_index))
```

Meaning: Absolute difference in average modulation burden between forward-moving and backward-moving beam groups.

beam_pair_balance_index

Beam-pair modulation balance

```
1 - abs(sum_forward(coupled_modulation_index) - sum_backward(coupled_modulation_index)) /  
(sum_forward + sum_backward)
```

Meaning: Balance score comparing the total modulation burden assigned to forward and backward beam groups.

small_opening_fraction

Fraction of narrow effective openings

```
weighted count(gap < 10 mm and gap > 0) / weighted count(gap > 0)
```

Meaning: Weighted fraction of effective openings smaller than 10 mm.

near_closed_fraction

Fraction of near-closed effective openings

```
weighted count(gap < 2 mm and gap > 0) / weighted count(gap > 0)
```

Meaning: Weighted fraction of effective openings smaller than 2 mm.

effective_small_gap_burden

Weighted burden of narrow effective gaps

```
weighted mean of max(0, 1-gap/10 mm) over positive zipped-channel gap proxies; abs(delta CMW)  
weights, zero increment falls back to 1
```

Meaning: Weighted severity score for narrow effective openings, with smaller openings contributing more burden.

projection_pitch_p95

95th percentile of projection pitch

```
nearest-rank 95th percentile of projection_pitch_i
```

Meaning: 95th percentile of projection pitch, using the nearest-rank interval summary.

projection_pitch_max

Maximum projection pitch

`max(projection_pitch_i)`

Meaning: Maximum projection pitch observed across adjacent control-point intervals.

projection_pitch_top3_mean

Mean of the top three projection-pitch intervals

mean of the three largest `projection_pitch_i` values

Meaning: Mean of the three largest projection-pitch intervals, or all intervals if fewer than three exist.

projection_mu_density_p95_proxy

95th percentile of the projection meterset-weight density proxy

nearest-rank 95th percentile of `projection_mu_density_proxy_i`

Meaning: 95th percentile of the projection MU-density proxy, using cumulative meterset-weight change per millimeter.

projection_mu_density_max_proxy

Maximum projection meterset-weight density proxy

`max(projection_mu_density_proxy_i)`

Meaning: Maximum projection MU-density proxy observed across adjacent control-point intervals.

projection_mu_density_top3_mean_proxy

Mean of the top three projection meterset-weight density proxy values

mean of the three largest `projection_mu_density_proxy_i` values

Meaning: Mean of the three largest projection MU-density proxy values.

projection_aperture_change_p95

95th percentile of the aperture-change rate

nearest-rank 95th percentile of `projection_aperture_change_i`

Meaning: 95th percentile of the projection aperture-change rate.

projection_aperture_change_max

Maximum aperture-change rate

`max(projection_aperture_change_i)`

Meaning: Maximum projection aperture-change rate observed across adjacent control-point intervals.

projection_aperture_change_top3_mean

Mean of the top three aperture-change rates

mean of the three largest `projection_aperture_change_i` values

Meaning: Mean of the three largest projection aperture-change rates.

projection_leaf_travel_p95

95th percentile of the leaf-travel rate

nearest-rank 95th percentile of `projection_leaf_travel_i`

Meaning: 95th percentile of the projection leaf-travel rate across both layers.

projection_leaf_travel_max

Maximum leaf-travel rate

`max(projection_leaf_travel_i)`

Meaning: Maximum projection leaf-travel rate across both layers.

projection_leaf_travel_top3_mean

Mean of the top three leaf-travel rates

mean of the three largest `projection_leaf_travel_i` values

Meaning: Mean of the three largest projection leaf-travel rates across both layers.

head_mu_density_mean_proxy

Head-third mean MU-density proxy

`mean(projection_mu_density_proxy_i)` over head-third intervals

Meaning: Mean projection MU-density proxy inside the head longitudinal third of the covered z-range.

head_mu_density_cv_proxy

Head-third MU-density proxy variability

`CV(projection_mu_density_proxy_i)` over head-third intervals

Meaning: Coefficient of variation of the projection MU-density proxy inside the head longitudinal third.

head_aperture_change_mean

Head-third mean aperture-change rate

`mean(projection_aperture_change_i)` over head-third intervals

Meaning: Mean projection aperture-change rate inside the head longitudinal third of the covered z-range.

head_aperture_change_cv

Head-third aperture-change variability

`CV(projection_aperture_change_i)` over head-third intervals

Meaning: Coefficient of variation of the projection aperture-change rate inside the head longitudinal third.

head_leaf_travel_mean

Head-third mean leaf-travel rate

`mean(projection_leaf_travel_i)` over head-third intervals

Meaning: Mean projection leaf-travel rate inside the head longitudinal third of the covered z-range.

head_leaf_travel_cv

Head-third leaf-travel variability

`CV(projection_leaf_travel_i)` over head-third intervals

Meaning: Coefficient of variation of the projection leaf-travel rate inside the head longitudinal third.

mid_mu_density_mean_proxy

Mid-third mean MU-density proxy

`mean(projection_mu_density_proxy_i)` over mid-third intervals

Meaning: Mean projection MU-density proxy inside the mid longitudinal third of the covered z-range.

mid_mu_density_cv_proxy

Mid-third MU-density proxy variability

`CV(projection_mu_density_proxy_i) over mid-third intervals`

Meaning: Coefficient of variation of the projection MU-density proxy inside the mid longitudinal third.

mid_aperture_change_mean

Mid-third mean aperture-change rate

`mean(projection_aperture_change_i) over mid-third intervals`

Meaning: Mean projection aperture-change rate inside the mid longitudinal third of the covered z-range.

mid_aperture_change_cv

Mid-third aperture-change variability

`CV(projection_aperture_change_i) over mid-third intervals`

Meaning: Coefficient of variation of the projection aperture-change rate inside the mid longitudinal third.

mid_leaf_travel_mean

Mid-third mean leaf-travel rate

`mean(projection_leaf_travel_i) over mid-third intervals`

Meaning: Mean projection leaf-travel rate inside the mid longitudinal third of the covered z-range.

mid_leaf_travel_cv

Mid-third leaf-travel variability

`CV(projection_leaf_travel_i) over mid-third intervals`

Meaning: Coefficient of variation of the projection leaf-travel rate inside the mid longitudinal third.

tail_mu_density_mean_proxy

Tail-third mean MU-density proxy

`mean(projection_mu_density_proxy_i) over tail-third intervals`

Meaning: Mean projection MU-density proxy inside the tail longitudinal third of the covered z-range.

tail_mu_density_cv_proxy

Tail-third MU-density proxy variability

`CV(projection_mu_density_proxy_i) over tail-third intervals`

Meaning: Coefficient of variation of the projection MU-density proxy inside the tail longitudinal third.

tail_aperture_change_mean

Tail-third mean aperture-change rate

`mean(projection_aperture_change_i) over tail-third intervals`

Meaning: Mean projection aperture-change rate inside the tail longitudinal third of the covered z-range.

tail_aperture_change_cv

Tail-third aperture-change variability

`CV(projection_aperture_change_i) over tail-third intervals`

Meaning: Coefficient of variation of the projection aperture-change rate inside the tail longitudinal third.

tail_leaf_travel_mean

Tail-third mean leaf-travel rate

`mean(projection_leaf_travel_i) over tail-third intervals`

Meaning: Mean projection leaf-travel rate inside the tail longitudinal third of the covered z-range.

tail_leaf_travel_cv

Tail-third leaf-travel variability

`CV(projection_leaf_travel_i) over tail-third intervals`

Meaning: Coefficient of variation of the projection leaf-travel rate inside the tail longitudinal third.

layer_imbalance_index

Dual-layer travel imbalance

`abs(mean(leaf_travel_i_MLCX1) - mean(leaf_travel_i_MLCX2)) / (abs(mean_MLCX1) + abs(mean_MLCX2))`

Meaning: Normalized difference between the mean MLCX1 and MLCX2 leaf-travel rates.

layer_correlation_index

Dual-layer travel correlation

`correlation(projection_leaf_travel_i_MLCX1, projection_leaf_travel_i_MLCX2)`

Meaning: Correlation between the interval-by-interval MLCX1 and MLCX2 leaf-travel rates.

x1_x2_aperture_disparity

Dual-layer aperture disparity surrogate

`mean over control points of abs(sum|MLCX1| - sum|MLCX2|) / (sum|MLCX1| + sum|MLCX2|)`

Meaning: Surrogate side-opening disparity, computed from the relative difference in MLCX1 and MLCX2 position magnitudes.

x1_x2_leaf_travel_ratio

Dual-layer leaf-travel ratio

`mean(projection_leaf_travel_i_MLCX1) / mean(projection_leaf_travel_i_MLCX2)`

Meaning: Ratio of mean MLCX1 leaf-travel rate to mean MLCX2 leaf-travel rate.

Section 3. V4 Physical Aperture Metrics

Version: aurora-v4-physical-aperture. Real leaf boundaries define a common y-sub-strip grid. Each layer's position vector is split into A/B banks; intervals are intersected across layers and clipped to jaws. A and P are exact area and union perimeter. These seven descriptors are a physical-geometry extension with adapted MCS, not exact McNiven MCS or a reproduction of the old unit-width design. Only positive cumulative-weight increments contribute, represented by endpoint i+1. Weights are delta CMW / beam final CMW times BeamMeterset. If any BeamMeterset is absent, every beam uses unit-normalized relative weights and a warning. Missing geometry or malformed/decreasing weights makes all seven unavailable. A closed positive-weight endpoint contributes BA/SAS/MCS zero and complexity one; BI/CA become unavailable for the entire aggregate, without dropping that endpoint.

mean_ba

mean_ba

`MUmean(A_i); A_i=sum_s h_i,s*g_i,s, exact physical dual-layer intersection area`

Meaning: Delivered-weight mean physical beam aperture area after intersecting both MLC layers and clipping to X/Y jaws using real leaf boundaries.

mean_bi

mean_bi

`MUmean(P_i^2/(4*pi*A_i)); P_i is the exact aperture-union perimeter; A_i=0 => unavailable`

Meaning: Delivered-weight mean boundary irregularity $P^2 / (4 \pi A)$, from the exact physical aperture union; unavailable if any positive-weight aperture is closed.

mean_ca

mean_ca

`MUmean(P_i/A_i); units mm^-1; A_i=0 => unavailable; this is not circularity`

Meaning: Delivered-weight mean perimeter-to-area ratio P/A in inverse millimeters; this CA is not circularity. Closed positive-weight apertures make it unavailable.

mean_sas5

mean_sas5

`MUmean(count_s(0<g_i,s<5 mm)/count_s(g_i,s>0)); positive-height sub-strips only; closed => 0`

Meaning: Delivered-weight mean fraction of positive open physical sub-strips with gap strictly below 5 mm; equal sub-strip counts, not area weighting; closed apertures contribute zero.

mean_sas10

mean_sas10

`MUmean(count_s(0<g_i,s<10 mm)/count_s(g_i,s>0)); positive-height sub-strips only; closed => 0`

Meaning: Delivered-weight mean fraction of positive open physical sub-strips with gap strictly below 10 mm; equal sub-strip counts, not area weighting; closed apertures contribute zero.

mean_mcs_aurora

mean_mcs_aurora

`MUmean(AAV_i*LSV_i); AAV_i=A_i/sum_s max_positive_endpoint(h_i,s*g_i,s);
LSV_i=1-sum_adj_open abs(delta g)/((n_open-1)*max_open(g)); n_open<=1 => LSV 1; zero
envelope => MCS 0`

Meaning: Delivered-weight mean AAV times gap-sequence LSV. AAV uses the beam sum of maximum clipped sub-strip areas at positive-weight endpoints. Adapted MCS, not exact McNiven MCS; differs from the old unit-width design.

mcs_complexity_aurora

mcs_complexity_aurora

1-mean_mcs_aurora; complement of the physical-aperture adapted MCS

Meaning: One minus the Aurora physical-aperture adapted MCS; larger values indicate greater modulation complexity under this version-specific definition.

Appendix A. Legacy Engineering Metrics

These legacy metrics are retained for engineering comparison with the first-pass Aurora prototype. They are not the preferred paper-style reporting set.

axial_travel_mm

Legacy total axial travel

$$\text{axial_travel_mm} = \sum_i |\text{delta } z_i|$$

Meaning: Legacy engineering metric for total absolute axial travel reconstructed from adjacent control-point isocenter z positions.

gantry_rotation_deg

Legacy total gantry rotation

$$\text{gantry_rotation_deg} = \sum_i |\text{delta } \theta_i|$$

Meaning: Legacy engineering metric for total absolute gantry rotation reconstructed from the observed angle sequence.

mm_per_deg

Legacy axial travel per gantry degree

$$\text{mm_per_deg} = \text{axial_travel_mm} / \text{gantry_rotation_deg}$$

Meaning: Legacy engineering metric for total axial travel divided by total gantry rotation.

mm_per_rotation

Legacy axial travel per full rotation

$$\text{mm_per_rotation} = \text{axial_travel_mm} / (\text{gantry_rotation_deg} / 360)$$

Meaning: Legacy engineering metric for total axial travel per full 360-degree gantry rotation.

pitch_consistency

Legacy pitch consistency metric

$$\begin{aligned} \text{interval_pitch}_i &= |\text{delta } z_i| / |\text{delta } \theta_i| \\ \text{pitch_consistency} &= \text{CV}(\text{interval_pitch}_i) \end{aligned}$$

Meaning: Legacy engineering pitch variability metric, equivalent to the coefficient of variation of interval pitch.

mu_per_mm

Legacy MU density along the axial axis

$$\text{mu_per_mm} = \text{total_mu_or_weight_proxy} / \text{axial_travel_mm}$$

Meaning: Legacy engineering MU density per millimeter of axial travel. Uses explicit total MU when available, otherwise cumulative meterset-weight span as a transparent proxy.

aperture_change_per_mm

Legacy aperture-change density

$$\text{aperture_change_per_mm} = \sum_i \text{delta } A_i / \sum_i |\text{delta } z_i|$$

Meaning: Legacy engineering metric for total absolute aperture-width change normalized by total axial travel.

leaf_travel_per_mm

Legacy leaf-travel density

```
leaf_travel_per_mm = sum_i delta L_i / sum_i |delta z_i|
```

Meaning: Legacy engineering metric for total absolute leaf travel normalized by total axial travel.

coupled_modulation_index

Legacy engineering aggregate

```
n(x) = x / (1 + x)
coupled_modulation_index = mean( n(aperture_change_per_mm), n(leaf_travel_per_mm),
n(mu_density_variability), n(pitch_consistency) )
```

Meaning: Legacy engineering aggregate of normalized aperture change, leaf travel, MU-density variability, and pitch variability.